

DRAWING PLOTTED:

STREAM NUMBER		1	2	3	4	5	6	7	8	9	10	11	12	13	14
COMPONENT															
SO2	SCFM	0	0	0	0	12,401	11,781	620	11,781	12,401	4,678	4,678	1,335	1,335	480
SO3	SCFM	0	0	0	0	227	216	11	216	227	7,950	7,950	11,293	11,293	12,148
O2	SCFM	23,003	23,003	23,003	23,003	10,261	9,748	513	9,748	10,261	6,400	6,400	4,728	4,728	4,300
N2	SCFM	86,808	86,808	86,808	86,808	86,808	82,468	4,340	82,468	86,808	86,808	86,808	86,808	86,808	86,808
DRY TOTAL	SCFM	109,811	109,811	109,811	109,811	109,697	104,212	5,485	104,212	109,697	105,836	105,836	104,164	104,164	103,736
H2O	SCFM	1,999	1,999	0	0	0	0	0	0	0	0	0	0	0	0
WET TOTAL	SCFM	111,809	111,809	109,811	109,811	109,697	104,212	5,485	104,212	109,697	105,836	105,836	104,164	104,164	103,736
PRESSURE	IN W.C.	0	-3	-13	205	196	196	196	180	179	145	136	131	119	113
TEMPERATURE	°F	93	93	150	263	2,080	2,080	2,080	705	779	1,144	806	965	806	847

STREAM NUMBER		15	16	17	18	19	20	21	22	23	24	25	26	27
COMPONENT														
SO2	SCFM	480	480	480	480	480	18	18	18	18	18	0.62	0	0
SO3	SCFM	12,148	12,148	0	0	0	462	462	462	462	0	0	0	0
O2	SCFM	4,300	4,300	4,300	4,300	4,300	4,069	4,069	4,069	4,069	4,069	4,175	115	115
N2	SCFM	86,808	86,808	86,808	86,808	86,808	86,808	86,808	86,808	86,808	86,808	87,240	432	432
DRY TOTAL	SCFM	103,736	103,736	91,588	91,588	91,588	91,357	91,357	91,357	91,357	90,896	91,416	547	547
H2O	SCFM	0	0	0	0	0	0	0	0	0	0	3,425	10	41
WET TOTAL	SCFM	103,736	103,736	91,588	91,588	91,588	91,357	91,357	91,357	91,357	90,896	94,842	557	588
PRESSURE	IN W.C.	101	92	69	61	52	47	39	36	32	17	0	0	343
TEMPERATURE	°F	548	330	180	571	779	808	638	461	275	172	76	93	120

STREAM NUMBER		50	300	301	302	303	304	305	306	307	308	309	310	311	312	313	314	315	316	317	318	319	320	700	701	702	703
COMPONENT																											
FLUID		Sulfur	98.50%	98.50%	98.50%	98.50%	98.50%	98.50%	98.50%	98.50%	98.50%	98.50%	98.50%	99.38%	98.50%	98.50%	98.50%	98.56%	98.50%	93.20%	93.20%	93.20%	93.20%	Water	Water	Water	Water
FLOW	LB/MIN	1,128	127,015	25,270	25,270	80,554	80,554	52,362	52,462	21,191	24,696	46,461	71,156	73,865	128	44,583	44,583	44,686	3,497	9,096	9,096	5,400	3,696	784	560	25	199
FLOW	GPM	75	8,662	1,723	1,705	5,494	5,384	3,500	3,509	1,445	1,651	3,150	4,800	5,090	8.7	3,015	3,000	3,020	234	612	602	357	245	94	67	3.0	24
TEMPERATURE	°F	275	214	214	180	214	150	150	153	214	150	196	180	244	187	188	172	186	150	160	104	104	104	66	66	66	66

STREAM NUMBER		600	601	602	603	610	620	621	622
COMPONENT									
FLUID		10% Na2SO4	10% Na2SO4	10% Na2SO4	10% Na2SO4	20% NaOH	Water	Water	Water
FLOW	LB/MIN	174,161	173,666	69	445	20	217	216	1.5
FLOW	GPM	19,556	19,500	7.8	50	1.9	26	26	0.2
TEMPERATURE	°F	76	76	76	76	66	66	66	66

STREAM NUMBER		900	901	902	903	904	905	906	907	908	909	910	911	912	913	914	915
COMPONENT																	
FLUID		Water	Water	Water	Water	Water	Water	Water	Water	Water	Water	Water	Water	Water	Water	Water	Water
FLOW	LB/MIN	21,676	11,900	11,900	1,747	1,747	4,185	4,185	3,845	3,845	21,676	128,300	113,000	113,000	15,300	15,300	128,300
FLOW	GPM	2,606	1,431	1,435	210	211	503	504	462	464	2,614	15,562	13,706	13,771	1,856	1,864	15,636
TEMPERATURE	°F	80	80	97	80	98	80	93	80	98	96.5	122	122	139	122	139	139

PRODUCTION: TOTAL PRODUCTION IS 2,480 STPD OF H2SO4 (100% BASIS)

PRODUCT ACID STRENGTH: 98.5 WT% (SUMMER), 93.2 WT% (WINTER)

GAS STRENGTH: 11.5 VOL% SO2 EQUIVALENT

EMISSIONS: 6.8 PPMV SO2 (DRY BASIS) EMITTED FROM STACK

STP: 32 °F, 1 ATM

AMBIENT MOISTURE: 0.0182 MOLE H2O / MOLE DRY AIR

BAROMETRIC PRESSURE: 0.84 ATM

PRESSURE VALUES ARE APPROXIMATE.

DIRTY CONDITIONS ARE SHOWN, WHICH INCLUDE THE FOLLOWING

ADDITIONAL PRESSURE DROPS:

30" W.C IN CONVERTER PASS 1

1" W.C IN THE INLET AIR FILTER

STREAM	PRESS.	TEMP	FLUID	NORMAL
NO.	PSIG	°F		FLOW
				LB/H
704	40	116	TREATED WATER + RETURN CONDENSATE	265,459
705	25	190	TREATED WATER + RETURN CONDENSATE	265,459
800	4.9	220	BOILER FEEDWATER	273,501
801	1,180	222	BOILER FEEDWATER	273,501
802	1,180	222	TG DESUPERHEATING WATER	3,687
803	987	223	BOILER FEEDWATER	269,814
804	969	295	BOILER FEEDWATER	269,814
805	951	401	BOILER FEEDWATER	269,814
806	933	465	BOILER FEEDWATER	269,814
807	915	536	HP SATURATED STEAM	264,418
808	900	608	HP SUPERHEATED STEAM	264,418
809	870	900	HP SUPERHEATED STEAM	264,418
810	933	538	HP BLOWDOWN	5,396
811	10	234	LP FLASH STEAM	1,873
812	10	234	LP BLOWDOWN	3,524
813	0	203	BLOWDOWN	3,412
814	0	203	VENT STEAM	112
815	70	324	LP STEAM IMPORT	6,670
816	10	283	LP STEAM TO DEAERATOR	8,542
817	5	220	VENT STEAM	500

1

MATERIAL AND ENERGY FLOW DIAGRAM DETAILS ARE FOR BASIC ENGINEERING DESIGN ONLY. ACTUAL PERFORMANCE MAY VARY DUE TO AMBIENT AND FEEDSTOCK CONDITIONS, MANUFACTURING TOLERANCE, SITE OPERATING CONDITIONS, EQUIPMENT DESIGN MARGINS IMPACT VALUES ON THIS DOCUMENT TO ADDRESS NORMAL, EXPECTED VARIATIONS DURING ACTUAL OPERATION.

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		NOTES:
1520-PR-PFD-0000-EXP-5005	COOLING WATER DISTRIBUTION	
1540-PR-PFD-0000-EXP-5004	BFW AND STREAM	
1530-PR-PFD-0000-EXP-5003	TAILGAS TREATMENT	
1520-PR-PFD-0000-EXP-5002	STRONG ACID	
1540-PR-PFD-0000-EXP-5001	GAS	
DWG. NO.	TITLE	
	REFERENCES	

					SEAL
1	RECONFIGURED FROM HRS READY TO DCDA				2024-09-23
0	ISSUED FOR DESIGN	RM	BA	TV	2024-01-16
C	REVISED DATA TABLES	RM	BA	TV	2023-12-20
B	ISSUED FOR CLIENT REVIEW	RM	BA	TV	2023-12-11
A	ISSUED FOR REVIEW	RM	BA		2023-11-23
NO.	DESCRIPTION	BY	CHK	APP'D	DATE
REVISIONS					

EXP
t: +1 713 439 3600 | f: +1 713 963 9085
exp.com

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	SCALE N/A	SIZE D	LithiumAmericas TITLE 2,250MTPD SA1 - DCDA MATERIAL BALANCE DESIGN CASE DIRTY	
	DRAWN BY R. MASANI	DATE 2024-01-16		
	CHECKED BY D. BIRD	DATE 2024-01-16		
	ENGINEERED BY B. AFELSKIE	DATE 2024-01-16	THACKER PASS PROJECT HUMBOLDT COUNTY, NV PROJ No. HOU-23001048-A1 REV	
APPROVED BY T. VAN DAELE		DATE 2024-01-16	DRAWING No. 1500-PR-HMB-0000-EXP-5006	
			1	

STREAM NUMBER		1	2	3	4	5	6	7	8	9	10	11	12	13	14
COMPONENT															
SO2	SCFM	0	0	0	0	13,020	12,369	651	12,369	13,020	4,900	4,900	1,416	1,416	518
SO3	SCFM	0	0	0	0	239	227	12	227	239	8,360	8,360	11,844	11,844	12,742
O2	SCFM	24,153	24,153	24,153	24,153	10,774	10,235	539	10,235	10,774	6,713	6,713	4,971	4,971	4,522
N2	SCFM	91,148	91,148	91,148	91,148	91,148	86,591	4,557	86,591	91,148	91,148	91,148	91,148	91,148	91,148
DRY TOTAL	SCFM	115,301	115,301	115,301	115,301	115,182	109,422	5,759	109,422	115,182	111,121	111,121	109,379	109,379	108,930
H2O	SCFM	2,098	2,098	0	0	0	0	0	0	0	0	0	0	0	0
WET TOTAL	SCFM	117,400	117,400	115,301	115,301	115,182	109,422	5,759	109,422	115,182	111,121	111,121	109,379	109,379	108,930
PRESSURE	IN W.C.	0	-2	-13	186	176	176	176	159	158	154	144	139	125	119
TEMPERATURE	°F	93	93	150	254	2,073	2,073	2,073	706	779	1,145	806	964	806	847

STREAM NUMBER		15	16	17	18	19	20	21	22	23	24	25	26	27
COMPONENT														
SO2	SCFM	518	518	518	518	518	19	19	19	19	19	0.65	0	0
SO3	SCFM	12,742	12,742	0	0	0	498	498	498	498	0	0	0	0
O2	SCFM	4,522	4,522	4,522	4,522	4,522	4,273	4,273	4,273	4,273	4,273	4,378	115	115
N2	SCFM	91,148	91,148	91,148	91,148	91,148	91,148	91,148	91,148	91,148	91,148	91,581	432	432
DRY TOTAL	SCFM	108,930	108,930	96,188	96,188	96,188	95,939	95,939	95,939	95,939	95,440	95,960	547	547
H2O	SCFM	0	0	0	0	0	0	0	0	0	0	3,594	10	41
WET TOTAL	SCFM	108,930	108,930	96,188	96,188	96,188	95,939	95,939	95,939	95,939	95,440	99,554	557	588
PRESSURE	IN W.C.	106	96	73	64	54	49	40	37	33	17	0	0	343
TEMPERATURE	°F	547	330	180	573	779	809	641	460	275	172	76	93	120

STREAM NUMBER		50	300	301	302	303	304	305	306	307	308	309	310	311	312	313	314	315	316	317	318	319	320	700	701	702	703
COMPONENT																											
FLUID		Sulfur	98.50%	98.50%	98.50%	98.50%	98.50%	98.50%	98.30%	98.50%	98.50%	98.50%	98.50%	99.42%	98.50%	98.50%	98.50%	98.56%	98.50%	93.20%	93.20%	93.20%	93.20%	Water	Water	Water	Water
FLOW	LB/MIN	1,184	127,190	24,828	24,828	81,686	81,686	52,362	52,467	20,676	25,652	45,504	71,156	73,998	138	44,583	44,583	44,694	3,672	9,280	9,280	5,400	3,881	823	587	27	209
FLOW	GPM	79	8,680	1,694	1,675	5,575	5,460	3,500	3,510	1,411	1,715	3,086	4,800	5,105	9.3	3,015	3,000	3,022	245	625	614	357	257	99	70	3.2	25
TEMPERATURE	°F	275	216	216	180	216	150	150	153	216	150	197	180	247	188	188	172	187	150	162	104	104	104	66	66	66	66

STREAM NUMBER		600	601	602	603	610	620	621	622
COMPONENT									
FLUID		10% Na2SO4	10% Na2SO4	10% Na2SO4	10% Na2SO4	20% NaOH	Water	Water	Water
FLOW	LB/MIN	174,163	173,666	73	445	21	228	226	1.5
FLOW	GPM	19,556	19,500	8.2	50	2.0	27	27	0.2
TEMPERATURE	°F	76	76	76	76	66	66	66	66

STREAM NUMBER		900	901	902	903	904	905	906	907	908	909	910	911	912	913	914	915
COMPONENT																	
FLUID		Water	Water	Water	Water	Water	Water	Water	Water	Water	Water	Water	Water	Water	Water	Water	Water
FLOW	LB/MIN	21,676	11,900	11,900	1,747	1,747	4,185	4,185	3,845	3,845	21,676	128,300	113,000	113,000	15,300	15,300	128,300
FLOW	GPM	2,606	1,431	1,435	210	211	503	504	462	464	2,614	15,562	13,706	13,775	1,856	1,865	15,640
TEMPERATURE	°F	80	80	98	80	98	80	93	80	98	97.0	122	122	140	122	140	140

PRODUCTION: TOTAL PRODUCTION IS 2,604 STPD OF H2SO4 (100% BASIS)
PRODUCT ACID STRENGTH: 98.5 WT% (SUMMER), 93.2 WT% (WINTER)
GAS STRENGTH: 11.5 VOL% SO2 EQUIVALENT
EMISSIONS: 6.8 PPMV SO2 (DRY BASIS) EMITTED FROM STACK
EQUIVALENT TO 6.95 LBS/HR AND 30.4 TONS/YEAR
STP: 32 °F, 1 ATM

AMBIENT MOISTURE: 0.0182 MOLE H₂O / MOLE DRY AIR
BAROMETRIC PRESSURE: 0.84 ATM
PRESSURE VALUES ARE APPROXIMATE.
CLEAN CONDITIONS ARE SHOWN

STREAM	PRESS.	TEMP	FLUID	NORMAL
NO.	PSIG	°F		FLOW
				LB/H
704	40	118.27	TREATED WATER + RETURN CONDENSATE	277,705
705	25	190	TREATED WATER + RETURN CONDENSATE	277,705
800	4.9	220	BOILER FEEDWATER	286,118
801	1,180	222	BOILER FEEDWATER	286,118
802	1,180	222	TG DESUPERHEATING WATER	3,747
803	987	223	BOILER FEEDWATER	282,371
804	969	295	BOILER FEEDWATER	282,371
805	951	401	BOILER FEEDWATER	282,371
806	933	465	BOILER FEEDWATER	282,371
807	915	536	HP SATURATED STEAM	276,723
808	900	608	HP SUPERHEATED STEAM	276,723
809	870	900	HP SUPERHEATED STEAM	276,723
810	933	538	HP BLOWDOWN	5,647
811	10	234	LP FLASH STEAM	1,960
812	10	234	LP BLOWDOWN	3,688
813	0	203	BLOWDOWN	3,571
814	0	203	VENT STEAM	117
815	70	324	LP STEAM IMPORT	6,959
816	10	283	LP STEAM TO DEAERATOR	8,915
817	5	220	VENT STEAM	501

STREAM NUMBER		600	601	602	603	610	620	621	622
COMPONENT									
FLUID		10% Na2SO4	10% Na2SO4	10% Na2SO4	10% Na2SO4	20% NaOH	Water	Water	Water
FLOW	LB/MIN	174,163	173,666	73	445	21	228	226	1.5
FLOW	GPM	19,556	19,500	8.2	50	2.0	27	27	0.2
TEMPERATURE	°F	76	76	76	76	66	66	66	66

STREAM NUMBER		900	901	902	903	904	905	906	907	908	909	910	911	912	913	914	915
COMPONENT																	
FLUID		Water	Water	Water	Water	Water	Water	Water	Water	Water	Water	Water	Water	Water	Water	Water	Water
FLOW	LB/MIN	21,676	11,900	11,900	1,747	1,747	4,185	4,185	3,845	3,845	21,676	128,300	113,000	113,000	15,300	15,300	128,300
FLOW	GPM	2,606	1,431	1,435	210	211	503	504	462	464	2,614	15,562	13,706	13,775	1,856	1,865	15,640
TEMPERATURE	°F	80	80	98	80	98	80	93	80	98	97.0	122	122	140	122	140	140

2234 gpm each at
normal

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1520-PR-PFD-0000-EXP-5005	COOLING WATER DISTRIBUTION																	
1540-PR-PFD-0000-EXP-5004	BFW AND STREAM																	
1530-PR-PFD-0000-EXP-5003	TAILGAS TREATMENT																	
1520-PR-PFD-0000-EXP-5002	STRONG ACID																	
1540-PR-PFD-0000-EXP-5001	GAS																	
DWG. NO.	TITLE																	
REFERENCES																		
1	RECONFIGURED FROM HRS READY TO DCDA																	
0	ISSUED FOR DESIGN	RM	BA	TV														
C	REVISED DATA TABLES	RM	BA	TV														
B	ISSUED FOR CLIENT REVIEW	RM	BA	TV														
A	ISSUED FOR REVIEW	RM	BA															
NO.	DESCRIPTION	BY	CHK	APP'D	DATE													
REVISIONS										EXP t: +1.713.439.3600 f: +1.713.963.9085 exp.com								

CASE 1A: SAP 100, LNC 100 (Guarantee)			Gross Power: 25,163 kW			Consumed: 10,600 kW			Net Power: 14,563 kW																		
STREAM NUMBER		139	140	141	142	143	144	145	146	147	149	150	151	156	157	160	161	162	163	165	166	168	169	802	809	815	
FLOW	LB/H	0	0	261,362	172,723	172,723	0	88,639	91,567	82,607	172,723	0	0	3,055	3,814	172,723	172,723	0	0	759	2,928	0	2,290	3,687	264,418	6,670	
PRESSURE	PSIG	70	145	870	-9.40	-9.40	870	70	70	70	40	870	-9.40	870	145	-9.40	40	725	725	1,180	1,180	40	70	1,180	870	70	
TEMPERATURE	°F	324	374	900	140	140	900	385	324	324	142	900	838	900	374	140	142	143	143	222	222	66	324	222	900	324	
CASE 1B: SAP 100, LNC 115			Gross Power: 24,462 kW			Consumed: 10,560 kW			Net Power: 13,902 kW																		
STREAM NUMBER		139	140	141	142	143	144	145	146	147	149	150	151	156	157	160	161	162	163	165	166	168	169	802	809	815	
FLOW	LB/H	0	0	260,751	159,911	159,911	0	100,840	104,081	95,102	159,969	0	0	3,666	4,577	159,969	159,969	0	0	911	3,241	0	2,290	4,152	264,418	6,689	
PRESSURE	PSIG	70	145	870	-9.40	-9.40	870	70	70	70	40	870	-9.40	870	145	-9.40	40	725	725	1,180	1,180	40	70	1,180	870	70	
TEMPERATURE	°F	324	374	900	140	140	900	383	324	324	142	900	838	900	374	140	142	143	143	222	222	66	324	222	900	324	
Case 1C: SAP 105 clean, LNC 100			Gross Power: 26,670 kW			Consumed: 10,599 kW			Net Power: 16,071 kW																		
STREAM NUMBER		139	140	141	142	143	144	145	146	147	149	150	151	156	157	160	161	162	163	165	166	168	169	802	809	815	
FLOW	LB/H	0	0	273,667	184,799	184,799	0	88,868	91,856	82,607	184,799	0	0	3,055	3,814	184,799	184,799	0	0	759	2,988	0	2,290	3,747	276,723	6,959	
PRESSURE	PSIG	70	145	870	-9.26	-9.26	870	70	70	70	40	870	-9.40	870	145	-9.40	40	725	725	1,180	1,180	40	70	1,180	870	70	
TEMPERATURE	°F	324	374	900	142	142	900	386	324	324	143	900	838	900	374	140	143	144	144	222	222	66	324	222	900	324	
CASE 2A: SAP 40, LNC MIN			Gross Power: 9,213 kW			Consumed: 5,347 kW			Net Power: 3,866 kW																		
STREAM NUMBER		139	140	141	142	143	144	145	146	147	149	150	151	156	157	160	161	162	163	165	166	168	169	802	809	815	
FLOW	LB/H	0	0	93,192	93,192	93,192	4,931	0	6,237	1,167	93,188	0	0	0	0	93,188	93,188	0	0	0	1,306	0	2,290	1,306	98,123	2,780	
PRESSURE	PSIG	70	145	870	-9.40	-9.40	870	70	70	70	40	870	-9.40	870	145	-9.40	40	725	725	1,180	1,180	40	70	1,180	870	70	
TEMPERATURE	°F	324	374	900	140	140	900	314	324	324	142	900	838	900	225	140	142	143	143	222	222	66	324	222	900	324	
CASE 3B: SAP 100, LNC min			Gross Power: 30,029 kW			Consumed: 10,647 kW			Net Power: 19,383 kW																		
STREAM NUMBER		139	140	141	142	143	144	145	146	147	149	150	151	156	157	160	161	162	163	165	166	168	169	802	809	815	
FLOW	LB/H	0	0	264,133	251,861	251,861	0	12,272	12,678	3,814	251,861	0	0	284	354	251,861	251,861	0	0	70	406	0	2,290	476	264,418	6,574	
PRESSURE	PSIG	70	145	870	-8.65	-8.65	870	70	70	70	40	870	-9.40	870	145	-8.65	40	725	725	1,180	1,180	40	70	1,180	870	70	
TEMPERATURE	°F	324	374	900	149	149	900	385	324	324	150	900	838	900	374	149	150	151	151	222	222	66	324	222	900	324	
CASE 4: SAP 25, LNC 0			Gross Power: 4,087 kW			Consumed: 5,026 kW			Net Power: -940 kW																		
STREAM NUMBER		139	140	141	142	143	144	145	146	147	149	150	151	156	157	160	161	162	163	165	166	168	169	802	809	815	
FLOW	LB/H	0	0	54,912	54,912	54,912	3,366	0	4,047	0	54,878	0	0	0	0	54,878	54,878	0	0	0	681	0	2,290	681	58,278	1,757	
PRESSURE	PSIG	70	145	870	-9.40	-9.40	870	70	70	70	40	870	-9.40	870	145	-9.40	40	725	725	1,180	1,180	40	70	1,180	870	70	
TEMPERATURE	°F	324	374	900	147	147	900	314	421	421	143	900	838	900	225	140	143	144	144	222	222	66	421	222	900	421	
			Check temperature																								
Case 5: SAP 100, LNC 0 (Full Bypass)			Gross Power: 0 kW			Consumed: 10,647 kW			Net Power: -10,647 kW																		
STREAM NUMBER		139	140	141	142	143	144	145	146	147	149	150	151	156	157	160	161	162	163	165	166	168	169	802	809	815	
FLOW	LB/H	0	0	0	0	333,176	7,046	0	8,910	0	257,376	257,371	333,176	0	0	333,181	333,181	75,805	75,805	0	1,864	0	2,290	1,864	264,417	6,620	
PRESSURE	PSIG	70	145	870	-2.10	-2.10	870	70	70	70	40	870	-2.10	870	145	-2.10	40	725	725	1,180	1,180	40	70	1,180	870	70	
TEMPERATURE	°F	324	374	900	194	228	900	380	324	324	194	900	228	900	225	194	194	195	195	222	222	66	324	222	900	324	
Case 6: SAP 40, LNC 0 (Partial Bypass)			Gross Power: 0 kW			Consumed: 5,479 kW			Net Power: -5,479 kW																		
STREAM NUMBER		139	140	141	142	143	144	145	146	147	149	150	151	156	157	160	161	162	163	165	166	168	169	802	809	815	
FLOW	LB/H	0	0	0	0	120,533	4,002	0	5,063	0	94,158	94,121	120,533	0	0	120,570	120,570	26,412	26,412	0	1,061	0	2,290	1,061	98,123	2,773	
PRESSURE	PSIG	70	145	870	-9.40	-9.40	870	70	70	70	40	870	-9.40	870	145	-9.40	40	725	725	1,180	1,180	40	70	1,180	870	70	
TEMPERATURE	°F	324	374	900	140	221	900	380	323	323	140	900	221	900	225	140	140	141	141	222	222	66	323	222	900	323	
Case 7: SAP 110, LNC 115			Gross Power: 27,677 kW			Consumed: 11,298 kW			Net Power: 16,378 kW																		
STREAM NUMBER		139	140	141	142	143	144	145	146	147	149	150	151	156	157	160	161	162	163	165	166	168	169	802	809	815	
FLOW	LB/H	0	0	288,446	187,140	187,140	0	101,306	104,712	95,102	187,140	0	0	3,666	4,577	187,140	187,140	0	0	911	3,407	0	2,290	4,318	292,112	7,320	
PRESSURE	PSIG	70	145	870	-9.12	-9.12	870	70	70	70	40	870	-9.40	870	145	-9.12	40	725	725	1,180	1,180	40	70	1,180	870	70	
TEMPERATURE	°F	324	374	900	144	144	900	386	324	324	145	900	838	900	374	144	145	146	146	222	222	66	324	222	900	324	

STP: 32 °F, 1 ATM

BAROMETRIC PRESSURE: 0.84 ATM

PRESSURE VALUES ARE APPROXIMATE.

POWER VALUES ARE APPROXIMATE.

MATERIAL AND ENERGY FLOW DIAGRAM DETAILS ARE FOR BASIC ENGINEERING DESIGN ONLY. ACTUAL PERFORMANCE MAY VARY DUE TO AMBIENT AND FEEDSTOCK CONDITIONS, MANUFACTURING TOLERANCE, SITE OPERATING CONDITIONS. EQUIPMENT DESIGN MARGINS IMPACT VALUES ON THIS DOCUMENT TO ADDRESS NORMAL, EXPECTED VARIATIONS DURING ACTUAL OPERATION.

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[illegible]

DRAWING PLOTTED:

STREAM NUMBER		320	400	401	402	405	406	407	408
COMPONENT									
FLUID		93.20%	93.20%	93.20%	93.20%	98.50%	98.50%	98.50%	98.50%
FLOW	LB/MIN	3,881	3,881	4,394	4,394	3,312	3,312	NNF	NNF
FLOW	GPM	257	257	291	291	215	216	NNF	NNF
TEMPERATURE	°F	104	AMBIENT	AMBIENT	AMBIENT	50	60	60	60
PRESSURE	PSIG	89	123	25	25	65	55	40	25

STREAM NUMBER		48	49	50	169	170	171	172	175	177	178	179	182	183	188	189	290	291	298	818
COMPONENT																				
FLUID		Sulfur	Sulfur	Sulfur	Steam	Steam	Steam	Condensate	Steam	Condensate	Steam	Condensate	Steam	Condensate	Condensate	Condensate	Condensate	Condensate	Condensate	Water
FLOW	LB/H	330,478	71,054	71,054	2,290	802	500	500	1,488	1,488	82	82	220	220	193	1,516	56	526	4,097	3,571
FLOW	GPM	368	79	79	1,546	541	446	1	1,004	3	73	0	196	0	761	3	220	1	9	7
TEMPERATURE	°F	275	275	275	324	324	315	295	324	314	315	295	315	295	203	203	203	203	203	203
PRESSURE	PSIG	44	GRAVITY	145	70	70	50	50	70	70	50	50	50	50	0	40	0	60	60	60

STREAM NUMBER		910	915	920	921	922	900	909	930	931	932	933
COMPONENT												
FLUID		Water	Water	Water	Water	Water	Water	Water	Water	Vapor	Vapor	Water
FLOW	LB/MIN	128,300	128,300	128,300	132,149	3,849	21,676	21,676	393	352	1	39
FLOW	GPM	15,562	15,640	15,563	16,030	467	2,606	2,614	47			5
TEMPERATURE	°F	122	140	122	122	122	80	97	80	97	97	80
PRESSURE	PSIG	40	30	20	20	40	40	30	60			40

STREAM NUMBER		149	168	189	190	257	258	270	271	272	274	280	281	294	295	296	297	298	299	509	620	700	704	930	933
COMPONENT																									
FLUID		Water	Water	Water	Water	Water	Water	Water	Water	Water	Water	Water	Water	Water	Water	Water	Water	Water	Water	Water	Water	Water	Water	Water	Water
FLOW	LB/H	184,799	0	1,516	0	97,078	499,967	401,991	491	0	388,311	25,622	0	291,233	140,768	0	125,677	4,097	0	1,568	13,681	49,378	277,705	23,563	2,350
FLOW	GPM	376	0	3	0	194	1,001	804	1	0	777	51	0	583	282	0	251	9	0	3	27	99	561	47	5
TEMPERATURE	°F	143	66	203	203	66	72	66	80	66	66	80	80	66	66	143	66	203	66	80	66	66	118	80	80
PRESSURE	PSIG	40	40	40	60	60	60	60	60	60	60	60	60	60	40	40	40	60	40	60	60	40	40	60	60

STREAM NUMBER		139	140	141	142	143	144	145	146	147	149	150	151	156	157	160	161	162	163	165	166	168	169	190	802	809	815
COMPONENT																											
FLOW	LB/H	0	0	273,667	184,799	184,799	0	88,868	91,856	82,607	184,799	0	0	3,055	3,814	184,799	184,799	0	0	759	2,988	0	2,290	0	3,747	276,723	6,959
FLOW	GPM	0	0	28,308	2,444,174	2,444,168	0	65,401	61,813	55,764	351	0	0	331	1,397	351	351	0	0	2	6	0	1,546	0	8	28,639	4,503
TEMPERATURE	°F	314	362	900	142	142	900	386	324	324	143	900	838	900	374	140	143	144	144	222	222	66	324	203	222	900	324
PRESSURE	PSIG	70	145	870	-9	-9	870	70	70	70	40	870	-9	870	145	-9	40	725	725	1,180	1,180	40	70	60	1,180	870	70

STREAM NUMBER		509	510	511	513	514	515	610	665 665
COMPONENT									
FLUID		Water	50% NaOH	50% NaOH	20% NaOH	20% NaOH	20% NaOH	20% NaOH	20% NaOH
FLOW	LB/MIN	26	722	17	23	0	44	21	0 0.24
FLOW	GPM	3	57	1	2	0	4	2	0 0.02
TEMPERATURE	°F	80	80	80	116	116	116	116	116
PRESSURE	PSIG	60	43 19	43	43	43	43	43	43

STREAM NUMBER		271	272	514	602	642	648	649	658	659	661	662	664	665	667
COMPONENT															
FLUID		Water	Water	20% NaOH	10wt% Na2SO4	5 wt% Na2SO4 Batch Effluent	ACW Batch Unload	Na2SO4 Batch Unload	Surfactant	Continuous Effluent	Water	Water	5 wt% Na2SO3 Steady State	20% NaOH	30wt% H2O2 Batch
FLOW	LB/H	491	NNF	NNF	4,367	529	NNF	121,205	TBD	6,098	491	TBD	96.0	14.7	48.5
FLOW	GPM	0.1	NNF	NNF	8	0.9	NNF	200	0.0	12.0	0.1	0.0	0.2	0.02	0.09
TEMPERATURE	°F	78	66	104	76	102	76	76	0	76	78	0	102	78	78
PRESSURE	PSIG	30	60	30	32	30	30	30	0	30	30	0	30	30	30

PRODUCTION: TOTAL PRODUCTION IS 2,604 STPD OF H2SO4 (100% BASIS) PRODUCT ACID

STRENGTH: 98.5 WT% (SUMMER), 93.2 WT% (WINTER)

STP: 32 °F, 1 ATM

AMBIENT MOISTURE: 0.0182 MOLE H2O / MOLE DRY AIR

BAROMETRIC PRESSURE: 0.84 ATM

PRESSURE VALUES ARE APPROXIMATE.

STREAM NUMBER		650	651	652	653	329
COMPONENT						
SO2	SCFM	0.08	0.08	0.00	0.15	0.03
H2S	SCFM	0.06	0.06	0.00	0.12	0.02
O2	SCFM	90	323	17	430	430
N2	SCFM	341	1,219	63	1,622	1,622
Sulfur	SCFM	0.10	0.10	0.00	0.20	0.02
DRY TOTAL	SCFM	432	1,542	79	2,052	2,052
H2O	SCFM	10	36	2	48	186
WET TOTAL	SCFM	442	1,578	81	2,101	2,238
TEMPERATURE	°F	275	275	93	268	102
PRESSURE	IN W. C.	-5	-5	-5	-5	1
PRESSURE	PSIG	-0.18	-0.18	-0.18	-0.18	0.04

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1520-PR-PFD-0000-EXP-5028	EFFLUENT TANK
1510-PR-PFD-0000-EXP-5026	20% CAUSTIC SODA SUPPLY
1560-PR-PFD-0000-EXP-5020	TG CONDENSER
1560-PR-PFD-0000-EXP-5018	WATER TREATMENT
1550-PR-PFD-0000-EXP-5016	ACID PLANT COOLING TOWER
1520-PR-PFD-0000-EXP-5014	FIN FAN COOLING SYSTEM
1510-PR-PFD-0000-EXP-5012	SULFUR
1570-PR-PFD-0000-EXP-5010	PRODUCT ACID
DWG. NO.	TITLE
REFERENCES	

NOTES:

1. PROCESS VALUES ARE PRELIMINARY. VALUES TO BE CONFIRMED BY SELECTED EQUIPMENT VENDORS.

0	ISSUED FOR DESIGN				2024-10-25
B	ISSUED FOR CLIENT REVIEW	AS	BA	TV	2024-09-23
A	ISSUED FOR REVIEW	AS	BA		2024-09-20
NO.	DESCRIPTION	BY	CHK	APP'D	DATE
REVISIONS					

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SCALE	N/A	SIZE	D
DRAWN BY	A. SKRYPCHENKO	DATE	2024-10-25
CHECKED BY	D. BIRD	DATE	2024-10-25
ENGINEERED BY	B. AFELSKIE	DATE	2024-10-25
APPROVED BY	T. VAN DAELE	DATE	2024-10-25



LithiumAmericas

2,250MTPD SA1 - DCDA
MATERIAL BALANCE
DESIGN CASE
105% RATE - UTILITIES

THACKER PASS
PROJECT
HUMBOLDT COUNTY, NV

PROJ No. HOU-23001048-A1
DRAWING No. 1500-PR-HMB-0000-EXP-5101
REV 0